

Heart Disease Risk Prediction & Patient Clustering

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PROJECT DEFINITION

The data problem for my this project will be using the UCI Heart Disease Data dataset from the UCI data repository. The dataset contains 920 observations of patient health data and demographics.

For the project, I will be using the mentioned dataset to determine the effectiveness of various Machine Learning (ML) models in predicting heart disease. For the analysis, I will utilize Python. The code will be provided in a Python Notebook attached to the assignment.

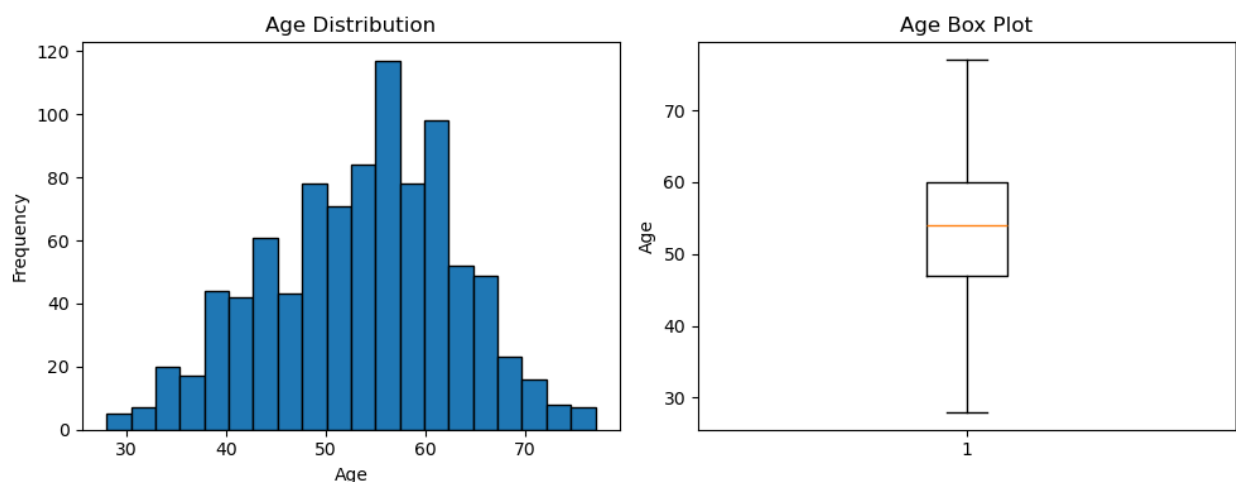
The main questions for the analysis are:

1. Can the variables in the UCI Heart Disease Data dataset be used to predict a positive heart disease diagnosis?
2. Are there natural patient groupings based on health metrics, and do these clusters align with heart disease diagnosis rates?

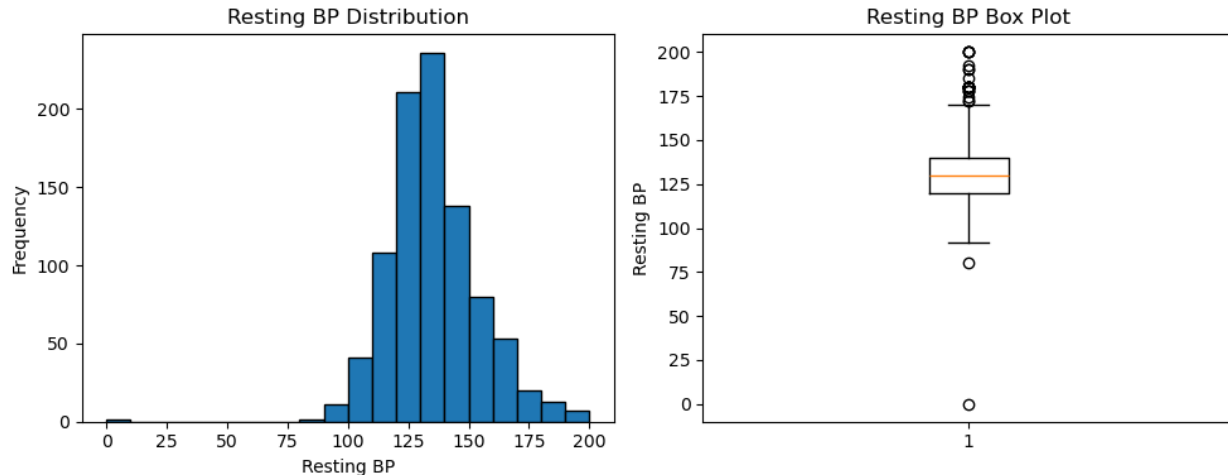
DATA COLLECTION AND CLEANING / EXPLORATORY DATA ANALYSIS (EDA)

For the data cleaning, I will move feature by feature. Any cleaning / transformations will be noted and a visualization of the distribution included.

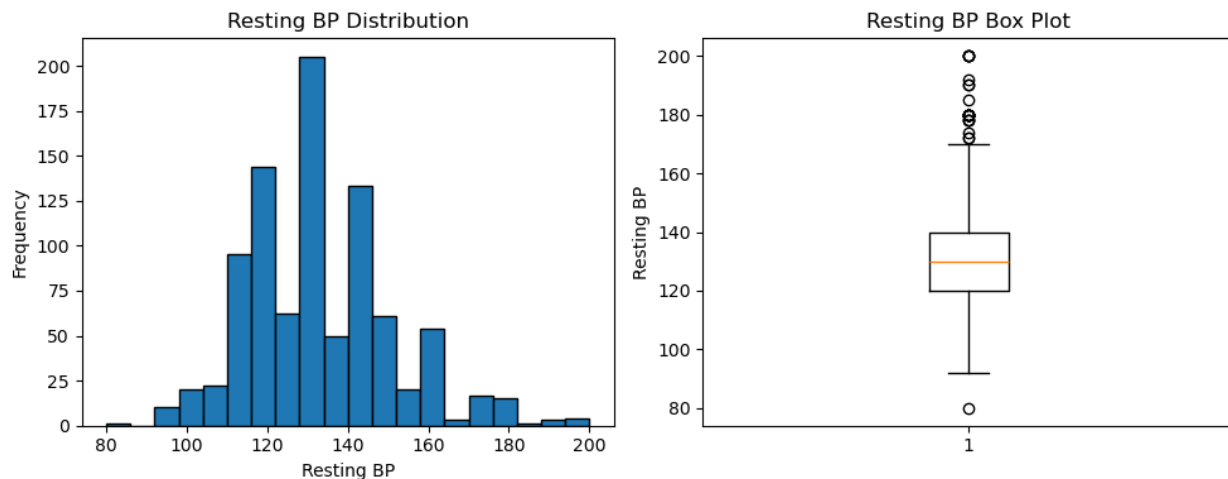
- age: The age feature, contained 920 of 920 instances and was normally distributed and no transformations were needed.



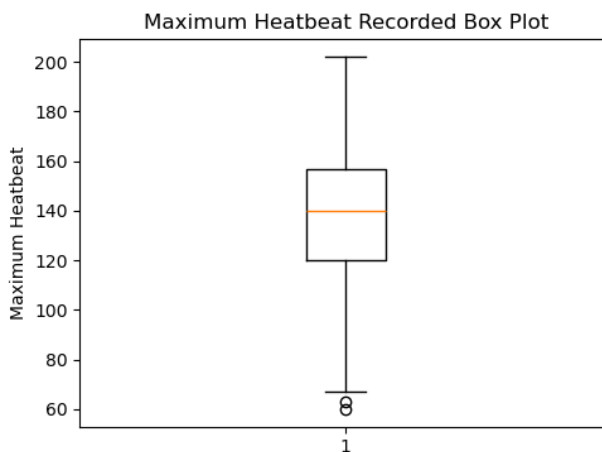
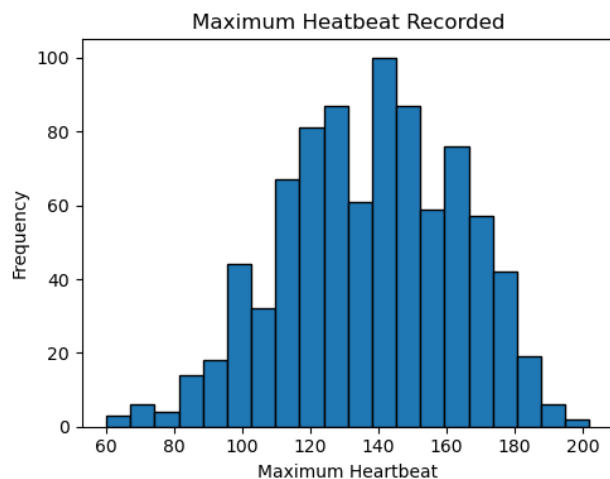
- `trestbps`: The Resting Blood Pressure (`trestbps`) contained 861 instances out of 920. There was one concerning outlier at a recorded Resting Blood Pressure of 0. Along with the one outlier, the remaining NaN data was replaced with the mean (132.28) of the feature. Provided below is the distribution before the transformation



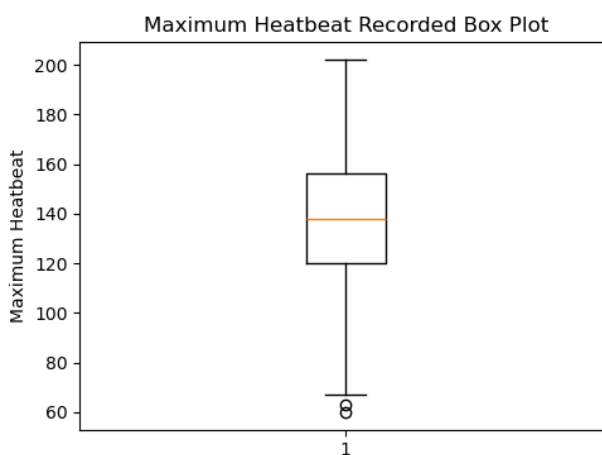
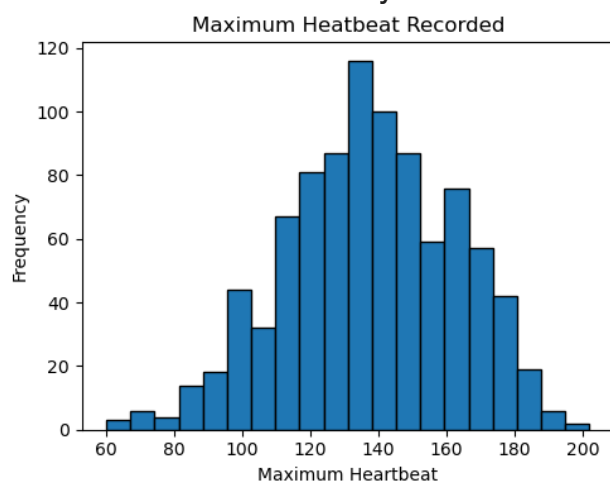
After the transformation, there are some outliers on the right side of the observations. However, with heart disease, non-normal blood-pressure recordings can be expected. Provided is the Resting BP Distribution (`trestbps`) after the transformation.



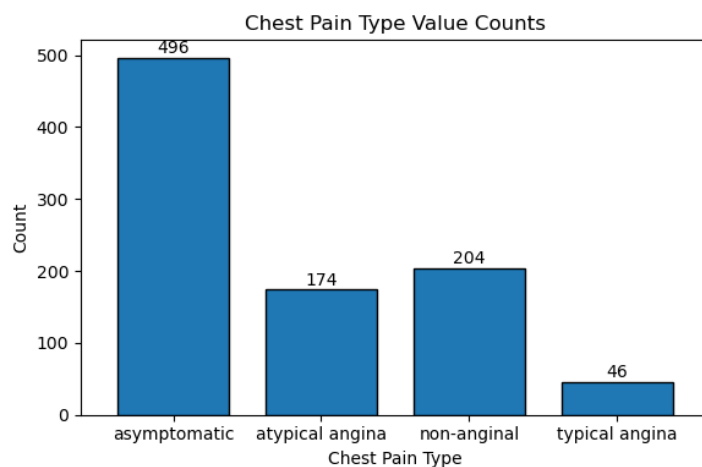
- The binary feature of a Fasting Blood Sugar > 120 (`fbs`) contained 692 FALSE and 128 TRUE observations. The remaining 90 NaN were replaced with the Mode value (FALSE).
- The Maximum Heartbeat Achieved (`thalch`) feature had a near-normal distribution with a couple outliers to the left of the distribution. There are 865 observations out of the 920 total.



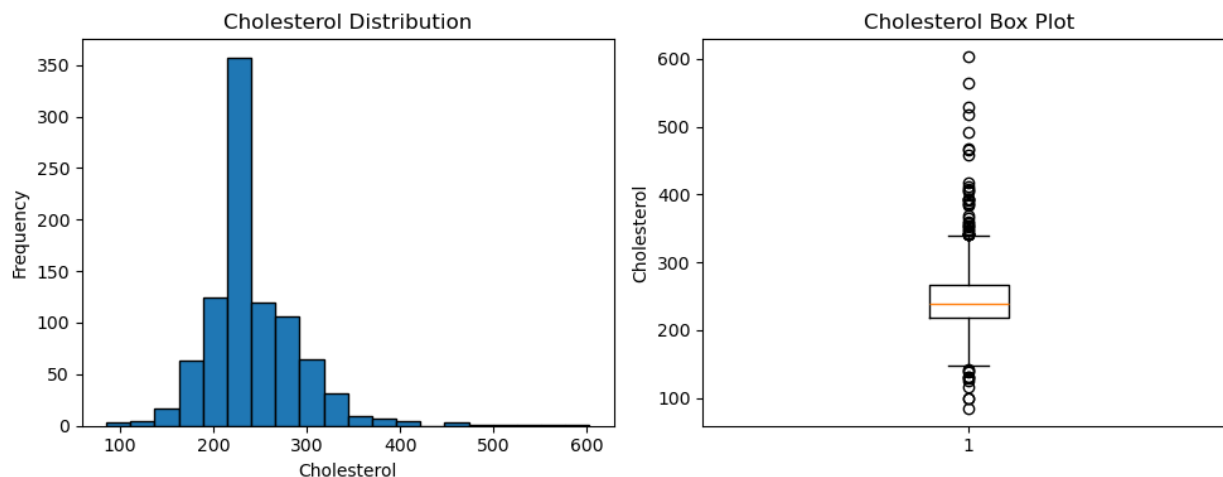
After replacing all NaN values with the mean Maximum Heartbeat (137.55), the distribution remained very near normal.



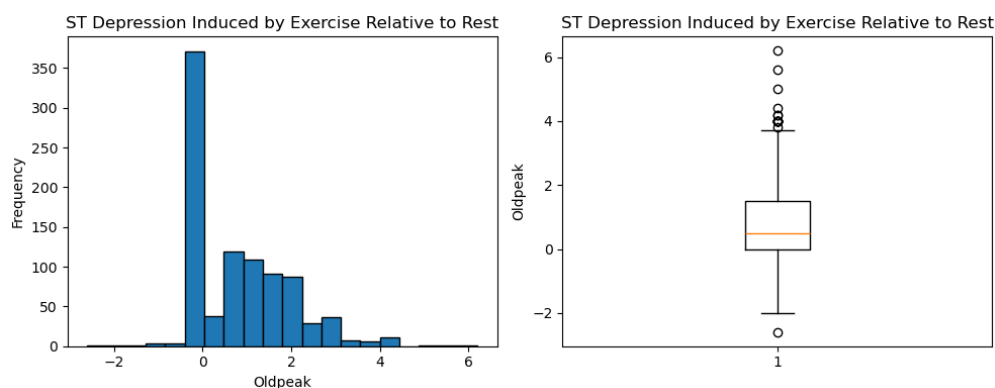
- The Chest Pain Type (cp) feature contained no NaN values. Provided is the breakdown of the types and their count.



- The Cholesterol Level in mg/dl (chol) contained 890 of the 920 observations. However, there were about 175 values that were 0. Again, this would appear to be an error in the test or record keeping. Those 0's taken out of the dataset left the Cholesterol feature skewed with a right tail. Therefore, the NaN data was replaced with the median (239.5), rather than the mean. The number of NaN values is not ideal, but the Cholesterol will likely be very important to the analysis and we do not want to lose 175 observations from the already small dataset.



- The Resting Echocardiogram Results (restecg) contained 918 observations of the 920. The feature was broken down into normal, st-t abnormality, and lv hypertrophy. The missing observations were replaced with the mode value (normal). After the transformation there were 188 lv hypertrophy, 179 st-t abnormality, and 553 normal observations.
- The Exercised Induced Angia binary column contained 865 observations out of the 920. The breakdown of 528 FALSE and 337 TRUE observations. After replacing the NaN observations with the mode (FALSE), there were 583 FALSE and 337 TRUE observations.
- The ST Depression Induced by Exercise Relative to Rest (oldpeak) contained 858 observations. The feature was skewed to the left and therefore a median (0.5) transformation was utilized.



- The following features had to be dropped because of excessing NaN observations:
 - Number of Major Vessels Colored by Fluoroscopy (66% NaN)
 - Slope of the Peak Exercise ST Segment (33.5% NaN)
 - Thal (53% NaN)

After transformations and data cleaning, we are left with 10 Feature columns and 1 Target column (num).

#	Column	Non-Null Count	Dtype
0	age	920 non-null	int64
1	sex	920 non-null	object
2	cp	920 non-null	object
3	trestbps	920 non-null	float64
4	chol	920 non-null	float64
5	fbs	920 non-null	float64
6	restecg	920 non-null	object
7	thalch	920 non-null	float64
8	exang	920 non-null	float64
9	oldpeak	920 non-null	float64
10	num	920 non-null	int64

FEATURE ENGINEERING

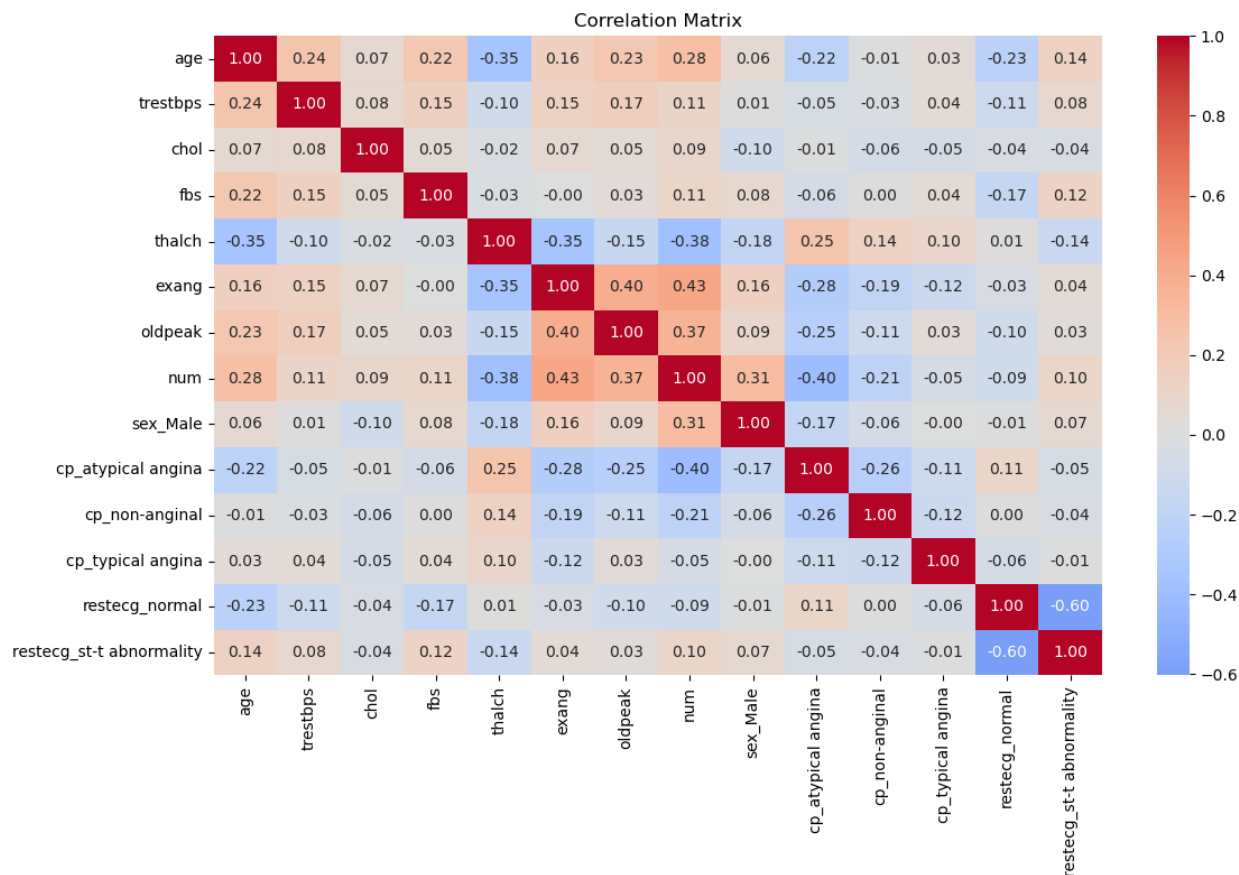
Beginning with feature engineering, the first step is to create dummy variables for all the object features [sex, cp, restecg]. The dummy variables will allow us to run tests with more models than without.

Additionally, the Heart Disease indication column is set at 1 for all values (levels of heart disease) greater than 1.

The data is split into 25% Test Data and 75% Training Data.

SMOTE is run on the Training Data to take the Training Data to a 50/50 rate on positive and negative heart disease diagnoses. The breakdown on the positive and negative heart rate diagnoses was 127 positive and 103 negative.

Before continuing with Model Selection, I ran a correlation heat map. I do not want to neglect multicollinearity within the data. Additionally, a heat map may provide a general idea of what variables appear to impact heart disease.



Based simply off the heat map, a higher max heart rate (thalch) is associated with a lower chance of heart disease, exercise induced angina (exang) is a positive predictor, ST depression (oldpeak) is also a positive predictor, and atypical angina is a negative predictor. There are other predictors as well, but these are the strongest (both positive and negative).

The heat map could potentially aid in decreasing the amount of variables used in the modeling of our data if the number of variables was an issue. In our case, there are not nearly enough variables/observations for this to be an issue.

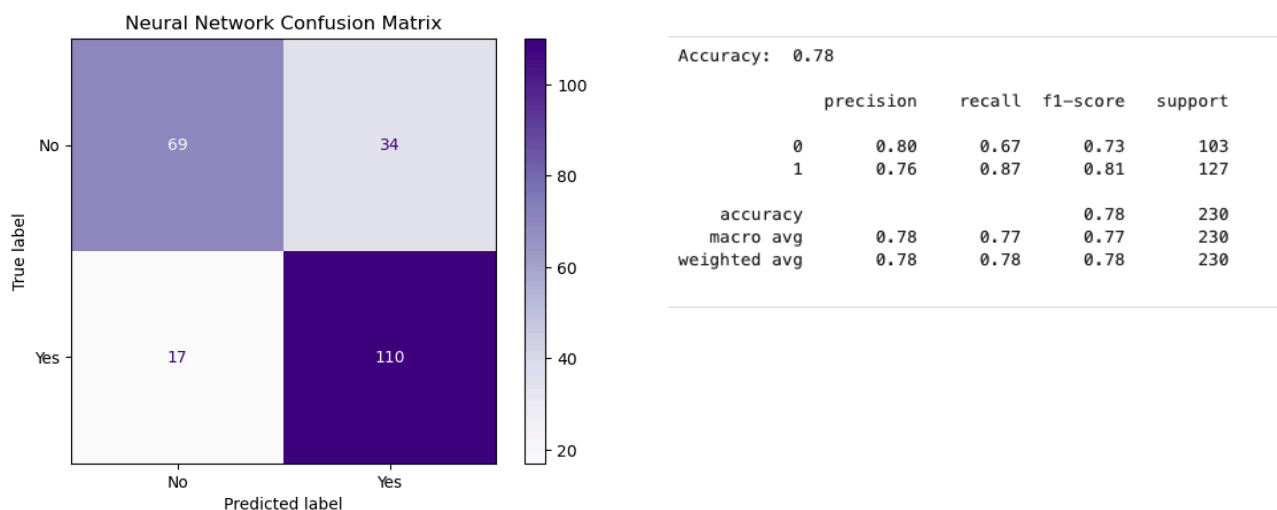
MODEL SELECTION

Supervised Model: For the supervised model, I will select a Neural Network. The Neural Network can capture any non-linear relationships within the data and give the flexibility of intensity with layers and nodes. A logistic regression is also a viable choice, but the non-normal chol and oldpeak can potentially cause issues in the model. The obvious weakness of the Neural Network is the black box nature of the model. However, we can certainly attempt to get the highest importances of the variables through a Permutation Importance test. The Supervised Model will give the best answer to the first business question.

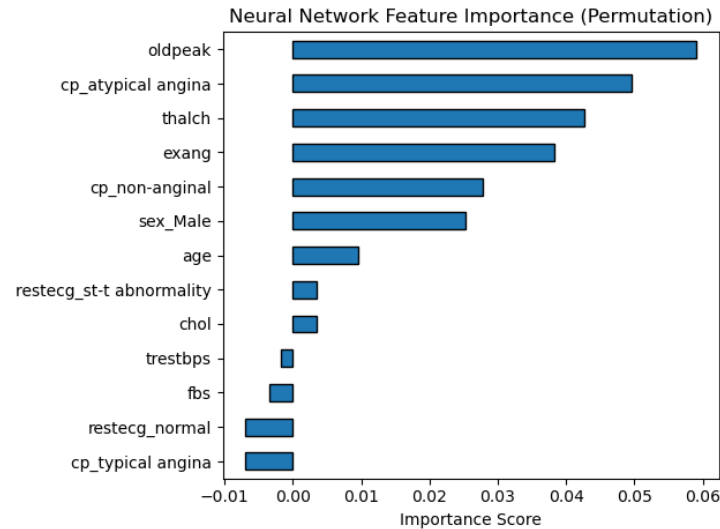
Unsupervised Model: For the second business question, a K-Means Clustering is selected. The K-Means is our best bet to see if there are any natural groupings within the data related to heart disease probabilities. K-Means helps by partitioning the data down to groups as best it can with similarity across the various features we have to help predict heart disease. We can then take the results of heart disease within the clusters to see what clusters indicate what level of risk for a heart disease diagnosis.

MODEL EVALUATION

Supervised Model (Neural Network): The Neural Network had an Accuracy score of 78%. With the amount of NaN data within the dataset, these are really great results. The model produces more False Positives than False Negatives. Ideally, the model would not produce any errors, but especially the False Negatives. While stressful, it is obviously more desirable to receive a False Positive and require more testing than a False Negative and be cleared while truly suffering from heart disease. The Precision and Recall are included below with the Confusion Matrix.

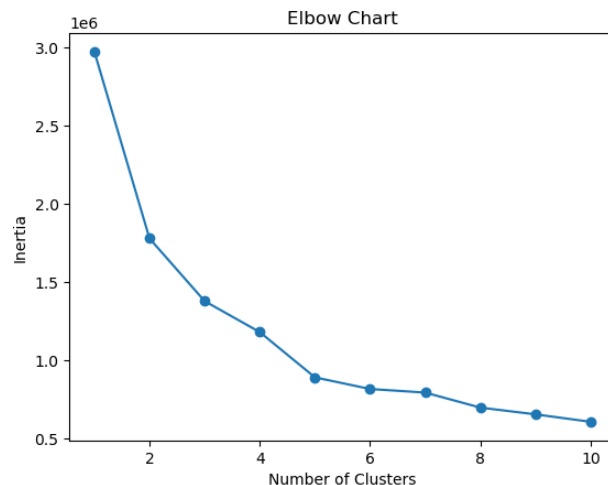


As mentioned in the Model Selection, the Neural Network is a black box model. However, the importances can be tested with a Permutation Test. The permutation test indicates the oldpeak to have the greatest importance score, followed closely by cp_atypical anger, thalch, and exang. Below are the results.



It would appear the ST Depression Induced by Exercise Relative to Rest (oldpeak) has the greatest impact. Looking back to the heat map, a greater oldpeak score positively correlates with a positive heart disease diagnosis. Likewise, a Chest Pain type of Typical Angina negatively correlates with a heart disease diagnosis. Maximum Heart Rate Achievable (thalach) also negatively correlates with a positive heart disease diagnosis. In this way, the heat map, combined with the permutation list give some indication of the factors the Neural Network is picking up on.

Unsupervised Model (K-Means Clustering): See the Elbow Chart for the K-Means Clustering below and Clustering Results below:



```
cluster
1    0.378601
0    0.550000
2    0.633147
Name: num, dtype: float64
```

cluster	num	count
0	1	77
	0	63
1	0	151
	1	92
2	1	340
	0	197

The Elbow indicates that 3 clusters appears to be the most optimal number. The results indicate three main categories for heart disease. Cluster 0 has a medium risk for heart disease (55%), Cluster 1 has the lowest risk (38%), and Cluster 2 has the greatest risk (63%). The results are broken down in a way that would be very useful for understanding one's risk of heart disease based on the features in this dataset. The clusters each have a significant number of observations continuing to make each

useful in better understanding if patients really can be broken down categorically into groups of various heart-disease risk based on the provided features.

Business Questions

As explained, a heart disease diagnosis can certainly be predicted with the data we had in this dataset. The data was also heavily altered by transformations due to missing observations, providing even more confidence if there was a more complete dataset. The permutation test, compared to the correlation heat map provided a better indication of how variables acted within the predictions of a heart disease diagnosis even though we were dealing with a Neural Network as our model.

Additionally, with this data, patients were grouped well within 3 clusters. The clusters allowed for a low, medium, and high risk status for a heart disease diagnosis. Answering the business question, patients can absolutely be grouped based on their feature similarity.

DOCUMENTATION

All documentation is provided in the attached Python Notebook.